

1. Which layer in a CNN is primarily responsible for reducing the spatial dimensions (width and height) of the input data?

- A) Convolutional Layer
- B) Activation Layer (ReLU)
- C) Pooling Layer (e.g., Max Pooling)
- D) Fully Connected Layer

Answer: C

2. When classifying environmental sounds, why is audio often converted into a Mel Spectrogram?

- A) To reduce the file size of the audio
- B) To transform the 1D audio signal into a 2D image-like representation for CNNs
- C) To remove background noise automatically
- D) To convert the audio into text

Answer: B

3. In Transfer Learning, what does "freezing" the base model layers (like MobileNetV2) achieve?

- A) It stops the model from making predictions
- B) It prevents the pre-trained weights from being updated during training
- C) It increases the learning rate automatically
- D) It deletes the weights to start from scratch

Answer: B

4. Which activation function is most commonly used in the final layer of a multi-class classification model to output probabilities?

- A) ReLU
- B) Sigmoid
- C) Tanh
- D) Softmax

Answer: D

5. What is the main advantage of MobileNetV2 over larger models like ResNet50 for a Smart Home Assistant?

- A) It has higher accuracy on all datasets
- B) It is lightweight and optimized for real-time performance on mobile/embedded devices
- C) It does not require any training data
- D) It only works with audio data

Answer: B

Part 2: Web Scraping & Pagination

6. Which pagination type allows you to "jump" directly to Page 10 by simply changing a number in the web address?

- A) Infinite Scroll
- B) Cursor-Based Pagination
- C) URL-Based Pagination
- D) "Load More" Buttons

Answer: C

7. Why is "Infinite Scroll" more difficult to scrape than standard numbered pagination?

- A) It uses higher encryption
- B) The URL stays the same, and content only loads when a scroll event is triggered
- C) It doesn't contain any HTML
- D) It requires a paid API key

Answer: B

8. In "Cursor-Based Pagination," what is a "cursor"?

- A) The mouse pointer on the screen
- B) A mathematical offset (e.g., \$+10\$)
- C) A unique ID or token that acts as a "bookmark" for the next set of data
- D) A JavaScript function that clicks buttons

Answer: C

9. Which tool is best suited for scraping a website that uses "Load More" buttons?

- A) BeautifulSoup (only)
- B) Requests (only)
- C) Selenium or Playwright
- D) Pandas

Answer: C

10. What is the main benefit of Client-Side Pagination for a user?

- A) It keeps the server from crashing
- B) It allows for instant page switching because all data is pre-loaded in the browser

- C) It hides the data from web scrapers
- D) It makes the initial page load much faster

Answer: B

Part 3: Flask & REST APIs

11. In your Flask code, what does jsonify(rates) do?

- A) It prints the rates to the console
- B) It converts a Python dictionary into a JSON-formatted response
- C) It saves the rates to a database
- D) It checks if the rates are valid numbers

Answer: B

12. Which HTTP method is typically used to "retrieve" data from a REST API endpoint like /rates?

- A) POST
- B) DELETE
- C) GET
- D) PUT

Answer: C

13. What happens if you run a Flask script and see ModuleNotFoundError: No module named 'flask'?

- A) Your code has a syntax error
- B) Flask is not installed in your current Python environment
- C) The server is already running on that port
- D) Your Python version is too old

Answer: B

14. In a REST API, what does the status code 404 represent?

- A) Unauthorized access
- B) Success
- C) Internal Server Error
- D) Resource Not Found

Answer: D

15. What is the purpose of the @app.route("/convert/<from_currency>/...") decorator?

- A) To encrypt the currency data
- B) To map a specific URL pattern to a Python function
- C) To create a new database table
- D) To install the Flask library

Answer: B